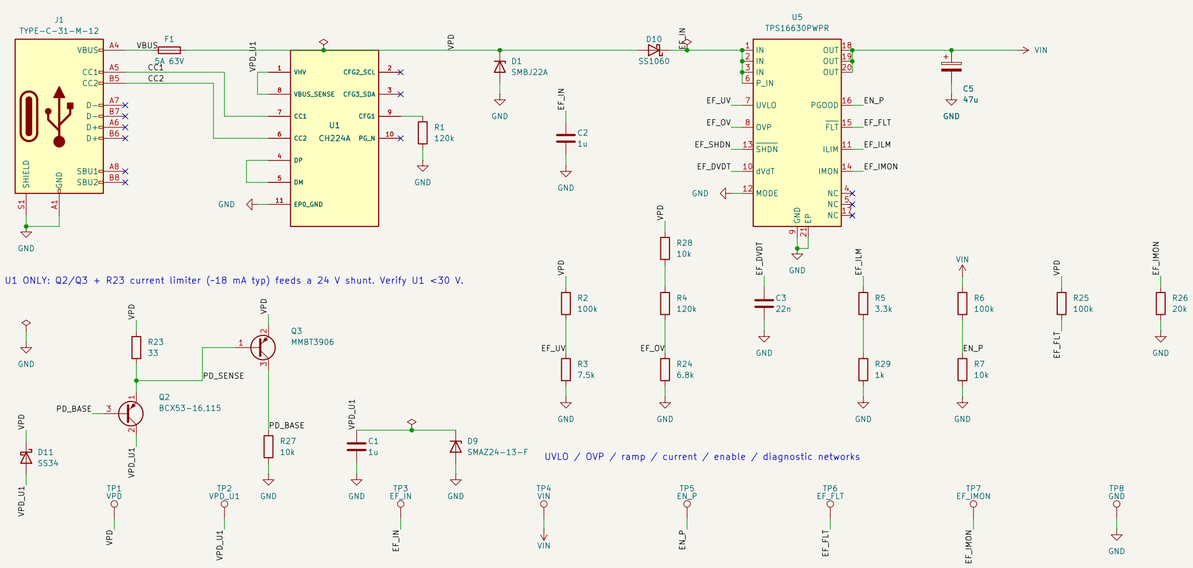
S5 INPUT: USB PD 20 V; 5 A contract/cable for full load. Board eFuse limit 4.19 A; no adapter-power tracking.

U5 TPS16630: 60 V input; UVLO 17.2 V / OVP 24.14 V nominal; ~9 ms ramp + PGOOD delay; auto-retry.

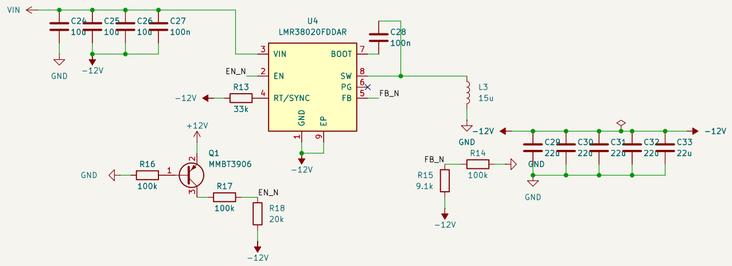


U1 ONLY: Q2/Q3 + R23 current limiter (~18 mA typ) feeds a 24 V shunt. Verify U1 <30 V.

TP5 = EN_P; TP6 = raw-voltage FLT; TP7 = 0.558 V/A typical; TP8 = system GND. D10 blocks USB backfeed.

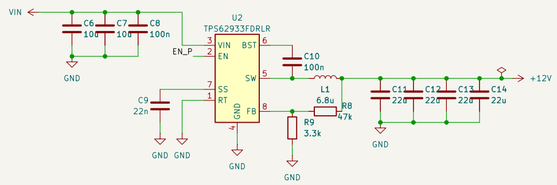
-12V: LMR38020F inverting buck-boost, forced PWM, ~790 kHz, 15 uH. IC ground = -12V.

Unchanged power stage. Q1 enables from +12V, not a precision all-rail supervisor. Validate loop stability.



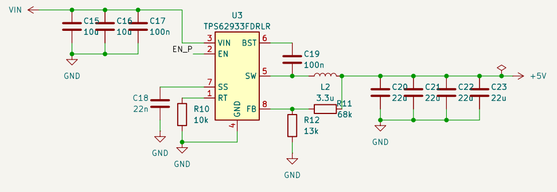
+12V: TPS62933F, forced PWM, 1.2 MHz, 6.8 uH

Unchanged divider: 0.8 x (1 + 47k/3.3k) = 12.19 V nominal

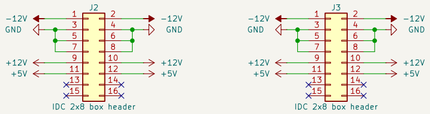


+5V: TPS62933F, forced PWM, ~2.0 MHz, 3.3 uH

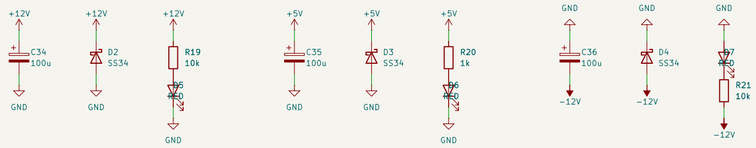
Unchanged divider: 0.8 x (1 + 68k/13k) = 4.98 V nominal



OUTPUTS: parallel 16-pin Eurorack headers (pin 1 = -12V). Not for paralleling independent supplies.

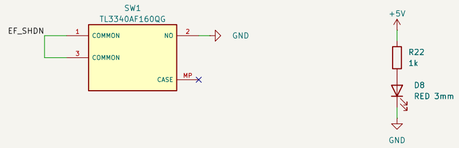


Bulk caps, polarity clamps and rail LEDs (unchanged)



PANEL: SW1 input disable/restart; D8 +5V presence

No guaranteed discharge time / complete module reset



TARGETS: +12V 2.5 A / +5V 2.5 A / -12V 1.0 A = 54.5 W total; simultaneous operation needs bench qualification.

Board limit: 4.3k = 3.3k + 1k; screened USB steady limit <=4.724 A. Full-load estimate ~66 W input, not a rating.

D10 upgraded to 10 A SMC; thermal copper required. Smaller sources may trip/cycle. Test thermal / PD / surge / startup.

Design notes and budget: docs/design.md

S5: board-side 4.19 A limit; 20 V / 5 A for full load; PGOOD; unchanged converters

Big Modular

Sheet: /

File: usb-power-45w.kicad_sch

Title: Big Modular — usb-power-45w

Size: A2	Date:	**Rev: S5**
KiCad E.D.A. 10.0.6		Id: 1/1